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Re: 90 FR 9088 Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Response prepared by Dylan Patel and AJ Kourabi.

Introduction

To win in AI, the US needs to win in AI infrastructure. We outline specific problems facing the US from winning the infrastructure race and key steps to overcome them.

We propose recommendations in four areas:

Summary of Key Recommendations

- (1) US-based chip manufacturing:
 - a. Heavily incentivize three key manufacturing capabilities: advanced logic, packaging, and HBM manufacturing.
 - b. Work across industry to enable the development and expansion of Intel Foundry's R&D.
 - c. Fast-track or exempt chip manufacturing facilities from onerous permitting requirements.
 - d. Set a national goal for domestic capacity to meet 50% of AI chip demand by 2028.
- (2) Energy supply for datacenters
 - a. Speed up grid interconnections (connecting local grids with power surplus to larger grids) for both power generators and power consumers and promote flexible next-generation solutions to improve overall grid utilization.
 - b. Simplify permitting datacenters to build their own power generation and natural gas infrastructure by reducing EPA regulatory requirements for air pollution permits.
 - c. Increase transparency on data center electricity use to strengthen market efficiency, improve pricing, and support investment growth.
- (3) Export Control Regulations
 - a. Expand controls to accommodate for interconnect bandwidth and networking technology.
 - b. Control the Nvidia H20 GPU to limit China's ability to develop advanced AI domestically.
 - c. Control semiconductor machine subsystems to prevent China from reengineering its way around controls.
- (4) Export Control Enforcement

- a. Increase funding and resources allocated to the Bureau of Industry and Security (BIS) for export enforcement.
- b. Pursue stronger coordination with allied countries on export controls to prevent stockpiling of restricted technologies.

AI is the critical technology change of the next few decades. We believe that AI will increasingly replace traditional software and expand capabilities in every aspect of technology and industry.

Traditional software has been labor-intensive but capital-light. Generative AI and large language models are not capital-light and represent a paradigm shift in infrastructure requirements.

An important observation in AI is Scaling Laws, which demonstrates that an increase in computing power, data, and parameters corresponds to an increase in capabilities for a model. The continuous increase in capabilities means that while AI may not be able to fulfill many aspects of economic activity today, continued progress in computing power should lead to a meaningful impact in almost all knowledge work in the future.

Like how most businesses were run not on the Internet pre-2000 and the following 20 years were about moving business and everyday functions to be Internet-enabled, AI will impact almost all information industries, such as biomedical engineering, manufacturing, software development, and military capabilities. The “digital transformation” will give way to AI transformation.

Most leading US technologies understand the critical importance of AI. Chinese counterparts such as DeepSeek and Alibaba understand that that large-scale infrastructure is a key competitive advantage in pursuing AI capabilities. Both American companies and Chinese counterparts are now engaging in a trillion-dollar arms race – and it is important the US government enables better and faster industrial production to support the United States companies and industries that are pursuing AI capabilities.

We believe that the US government can accomplish meaningful support to the burgeoning AI industry by following our four key suggestions. Starting with US manufacturing of the key element of AI: Compute.

1) US-Based Manufacturing

Today, we are at the lowest level of domestic semiconductor manufacturing in this country's history. Despite *inventing the semiconductor* and creating most of the value from its subsequent industrialization, the U.S. struggles to produce semiconductors domestically today. Taiwan is committed to ensuring that the leading technologies remains in Taiwan and not be moved over to the U.S., despite considerable subsidies, including Research and Development (R&D).

Supply of semiconductors must be guaranteed by onshoring manufacturing. Without capacity, all other capabilities pursued by the U.S. could be cut off very quickly. An analogy is owning power plants operating on natural gas but having to import natural gas. If we cannot produce the core supply needed for AI, it's hard for the U.S. to have AI self-sufficiency.

Were access to TSMC's Taiwan manufacturing capabilities cut off today, AI chip shipments would drop to zero. Even in the most optimistic timeline, it would take five years at minimum to build new U.S. manufacturing capacity to replace what Taiwan can build in 2025:

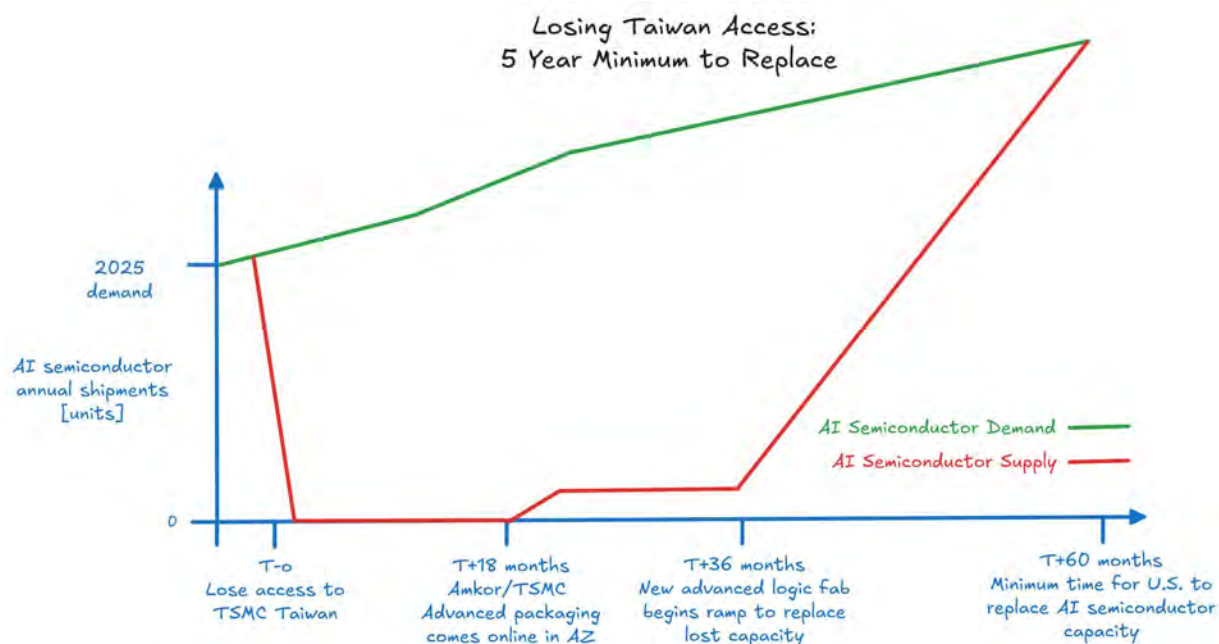


Figure 1 - Losing TSMC production today would create at least a five-year gap in AI chip supply—a blow to U.S. competitiveness. Source: SemiAnalysis

In the meantime, China would catch and surpass the Western world in advanced semiconductor manufacturing. While their capabilities today are ~5 years behind, the U.S. would lose advanced R&D along with Taiwanese manufacturing. There is no prospect of restoring leading-edge R&D capability from zero in a 5-year timespan. **It is critical that the U.S. maintain leading-edge semiconductor R&D.** The only U.S. entity capable of this today is Intel Foundry. Thus, it must be kept alive for economic and national security reasons.

Our analysis considers the multiple parts needed to make a leading high end AI chip today. There are separate manufacturing capabilities needed to produce these chips: **advanced logic for GPUs, DRAM for HBM** (high bandwidth memory), and **advanced packaging** to combine them.

Advanced Logic

Advanced logic refers to semiconductors fabricated at the smallest dimensions, often called the “leading edge.” In the context of GPUs or accelerators for AI, advanced logic refers to the brains of the operation. TSMC is the only viable producer of semiconductors for GPUs. Advanced logic manufacturing also has other applications including CPUs, switches, mobile semiconductors (system on chip, or SoC), and other logic. TSMC is based in Taiwan with only a few fabrication facilities in the US. This is a critical chokepoint for the United States.

Advanced logic is a bottleneck as TSMC Arizona can only supply **~10% of demand for AI**. This excludes semiconductor supply needed for phones, networking equipment, desktop computers and other critical capabilities. Advanced logic is the longest lead time to replacement, even with an emergency it would take 36 months minimum from project funding to first wafers out. Even with aggressive ramps into high volumes, it's unlikely we could replace demand in 5+ years of supply response. For the entire duration of the supply ramp, the United States would be severely crippled in AI chips, not to mention networking, phones, and computers.

This is the single greatest risk to America's technological future and security. To de-risk this critical chokepoint, manufacturers must be incentivized to build new capacity in the U.S. There are two key issues that make the U.S. less desirable than Asian countries for new fab buildouts: **cost** and **time-to-market**. Higher costs in the U.S. are often blamed on labor, but fabs are largely automated and operating expenses are largely driven by depreciation and raw materials, not personnel. Initial fab construction is more expensive in the United States, and we believe consists of up to 30-40% of the capital expense for building and outfitting a new facility.

The main cost difference is artificial, driven by subsidies. Foreign countries heavily subsidize chip manufacturing such that building and operating fabs is significantly cheaper (this excludes possible CHIPS Act disbursements). China provides more than \$15B in direct subsidies to its semiconductor manufacturing industry each year. Japan provides subsidies and other incentives totaling at least \$13B annually through 2030, and Taiwan credits up to 25% of chip R&D expenses.

Another cornerstone capacity is leading edge R&D, which must also be done in the U.S. The rapid pace of advanced logic development – characterized by Moore's Law – means that leading-edge fabs are superseded within 1-2 years. R&D is required to keep the pace of building capacity at the frontier. TSMC fabs built in the U.S. are just a temporary stopgap as they're still dependent on R&D in Taiwan. This is "leasing" the technology not owning it and creating true security.

The best option is likely Intel. Intel foundry has execution issues, but it is the best-positioned American company with both leading-edge R&D and high-volume manufacturing. Having either R&D or capacity does not create security, but Intel foundry is the most likely candidate to achieve both. A further collaboration between industry, Intel Foundry, and the government is needed to support and position Intel to meet this critical American need.

HBM (High Bandwidth Memory)

High Bandwidth Memory (HBM) is a key distinguishing feature of modern AI accelerators like GPUs and fabricated separately. As models scale to larger sizes, HBM is the critical piece that allows for accessing, processing, and moving large amounts of data for model training or inference. This is a signature design constraint in large language models, and HBM is the second most critical subcomponent.

The U.S. has no onshore HBM manufacturing. HBM requires DRAM (Dynamic Random Access Memory) and an advanced packaging facility to stack DRAM into a single HBM package. The CHIPS Act

incentivized a Micron DRAM plant and SK Hynix packaging fab, but these cannot use each other's products and, therefore, cannot produce HBM. Incentives need to focus on a single manufacturer onshoring both DRAM and HBM packaging.

Micron is the best candidate as it's the only U.S.-based company capable of producing HBM today. R&D is done at its U.S. facilities, but all mass production is in Asia. Given that DRAM is commoditized, cost is the key consideration for effective incentives to bring manufacturing onshore. Incentivizing and encouraging high volume manufacturing will be important, but cost consideration is the biggest hurdle, given memory is an intensely cost competitive business. Micron is capable of this but this needs to happen with urgency.

Advanced Packaging

Advanced Packaging is the important glue that connects the advanced logic and HBM together into a single package¹. Chip on Wafer on Substrate (CoWoS) is the key technology used today in packaging GPUs and accelerators by all leading players. CoWoS is a technology that requires specialized equipment to connect parts together with extreme precision and alignment for the total package to work together. TSMC is once again a leader in the space and has most of the capacity of this technology.

There is no advanced packaging capacity and barely any R&D in the United States today. Packaging technology was historically considered a "back end" and not a differentiated process, but today we believe it should be considered as important as the logic and HBM needs. We understand an Amkor/TSMC joint facility with this capability is slated for construction in Arizona but won't come online until mid-2026 at the earliest. This facility needs to be accelerated and expanded, and it's possible that Intel Foundry could pursue packaging capacity and R&D domestically to create semiconductor security.

Recommendations

The recent efforts in the United States have begun the lengthy and expensive process of bringing back semiconductor production to the United States. **It is not enough.** For the United States to lead in the AI technology race, it must lead in semiconductor manufacturing. We have the technological potential – but the intention and focus are insufficient. The United States must make US semiconductor manufacturing a top priority. We recommend:

- Encourage more high-volume manufacturing from TSMC and more R&D in the United States.
- Intel Foundry (or successor entity) must be the ultimate owner of high-volume manufacturing and R&D. A split as well as an industry consortium to closely partner Intel with actual customers to solve solutions is needed.

¹ Dylan Patel et al., *AI Capacity Constraints—CoWoS and HBM Supply Chain*, SemiAnalysis (July 5, 2023), Available at <https://semianalysis.com/2023/07/05/ai-capacity-constraints-cowos-and/>. (Accessed March 13, 2025).

- Accelerate and cheapen the time to build a fab in the U.S. through removing regulatory hurdles such as permitting and planning. Even the fastest buildouts can take 736 days from capital to production. Japan and Korea's timelines are 584 days and 620 days respectively.²
- Electricity is a key constraint. Federal and state governments can accelerate power timelines by fast-tracking interconnect approvals, aiming to reduce the year timeline to months.
- The United States needs a goal to reach for. **We believe over 50% of US AI chip demand should be domestically manufactured by 2028.** This means an ecosystem investment in advanced logic, HBM, and advanced packaging.

In sum, TSMC's U.S. fab meets only 10% of AI chip needs, Micron has no domestic HBM production, and advanced packaging capacity is almost entirely offshore. Without urgent interventions, these gaps will widen in the coming years.

2) Energy Supply for Datacenters

AI models demand vast amounts of data and compute, thus requiring meaningful amounts of power for training and inferencing AI models. As users continue to adopt models, this means more power for AI accelerators, such as GPUs, to train and infer the models. Users are growing, and with improving technology means power is growing.

The physical location of where the compute for AI happens is in the datacenter. The datacenter houses the servers and networking used to train and serve models to users. Because of the increasing demands from users and models, datacenters are struggling to keep up. This is partially because of the United States' lagging energy infrastructure. The last decade of power usage in America has been stagnant, while businesses have become more efficient in creating economic value with the same amount of power.

AI changes this. America's aging and slow power grid risks slowing down the AI infrastructure buildout – benefitting rival countries like China with more flexibility and agility. Building the datacenters themselves is also subject to heavy permitting overhead.

SemiAnalysis forecasts AI Energy demand to reach close to **150GW** of Critical IT power by 2030. Hosting 50% of this in the US would imply a **~10% growth to the US peak electricity demand** (745GW) in five years – after **15 years with near zero electricity demand growth** in the United States. Additionally, we forecast that global GPU manufacturing capacity will significantly outpace datacenter construction. Our analysis indicates that by the end of this year, there will be approximately 40GWs more power demand from chips than worldwide datacenter projects. The key question is how much of those 40GW will be

² John VerWey, *No Permits, No Fabs: The Importance of Regulatory Permitting for Semiconductor Manufacturing*, Ctr. for Sec. & Emerging Tech. (Oct. 2021), Available at: <https://cset.georgetown.edu/wp-content/uploads/CSET-No-Permits-No-Fabs.pdf>. (Accessed Mar. 11, 2025)

hosted in the US or in other nations. If power generation does not increase, the United States will fall behind.

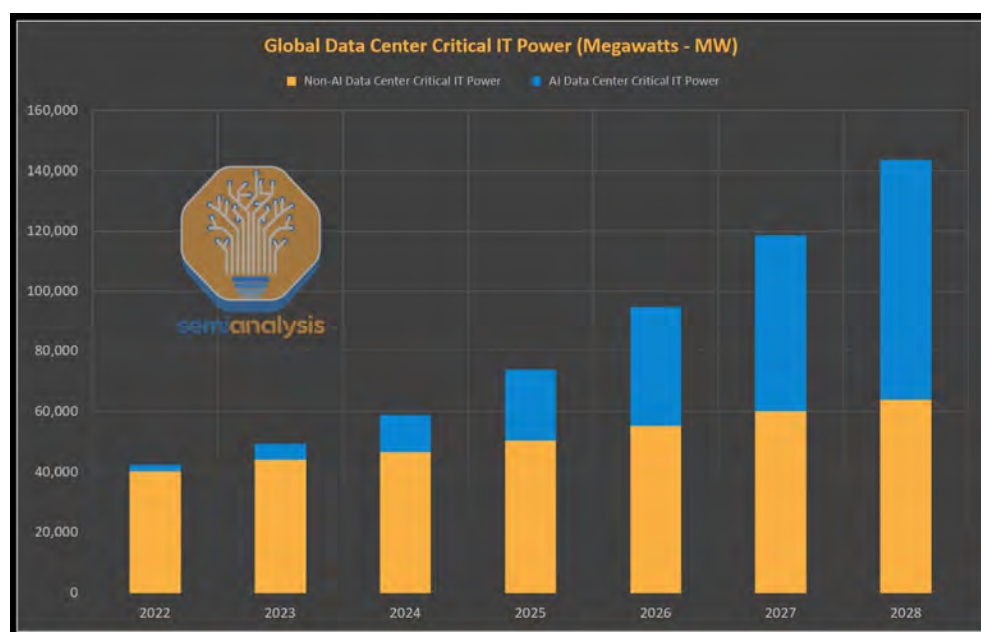


Figure 2 -Datacenter Critical IT Power. Source: SemiAnalysis

This is critical to winning the future of AI. We propose a simple framework to accelerate AI power adoption in the United States. We must **accelerate grid interconnection, improve on-site power generation, and increase information for datacenter** participants.

Accelerate the Grid Interconnection Process

Our conservative US datacenter growth scenario still dwarfs the total power generation to be added to the US power grid. We estimate a gap **north of 50GW by 2028**. This is a result of a timing mismatch: while AI datacenters prioritize time-to-market, adding power generation is anything but quick – wait times on average are over five years in the “project queue” before it can deliver power and serve users, as indicated by a Department of Energy report funded under contract DE-AC02-05CH11231³.

In some areas in the country, underutilized power assets enable datacenters to be served without building new infrastructure. But they also face a “load interconnection” process and enter a queue of projects – which we estimate at **north of 350GW** currently in the US. AI datacenter developers are frantically searching for power, and transmission utilities are overwhelmed.

Viable solutions are not being pursued today. Interconnection can happen faster by incentivizing utilities to explore innovative solutions like demand-response programs via batteries or backup generators,

³ Joseph Rand et al., *Queued Up: Characteristics of Power Plants Seeking Transmission Interconnection As of the End of 2023*, Lawrence Berkeley Nat'l Lab (Apr. 2024), Available at: https://emp.lbl.gov/sites/default/files/2024-04/Queued%20Up%202024%20Edition_1.pdf. (Accessed Mar. 6, 2025)

increasing utilization rates of existing transmission assets, or building new transmission assets. All of these would increase utilization of the current energy infrastructure.

The Department of Energy and the Federal Energy Regulatory Commission can implement solutions that have already been proposed, such as research from the Nicholas Institute that demonstrates how the grid could support significantly more load through minimal curtailment⁴.

Accelerate On-site Power Generation

Leading AI-focused corporations like xAI have shifted to on-site power generation to bypass the energy bottleneck. This significantly lowers the constraint on nationwide transmission and generation assets and accelerates time-to-market for AI datacenters.

Unfortunately, challenges are slowing down the adoption of such systems:

- Heavy environmental permitting constraints slow down natural gas generation onsite.
- Developers prefer a grid connection to provide backup power - and ISOs and RTOs struggle to account for such configurations, slowing down the approval process.

Most utilities do not have experience in large load customers with on-site power generation but a grid connection for backup power. In some areas, the regulatory framework doesn't allow such arrangements. This leads to a slowdown in the development of large datacenters with on-site power generation.

The Executive Branch should consider a national framework for utilities to evaluate large loads with on-site power generation. The framework should provide multiple options to serve a broad range of target datacenter uptime, for example:

- Electrical grid is a secondary flexible power, but not a backup for a primary source, such as the AEP Ohio arrangement with natural gas fuel cells acting as primary power.
- Electrical Grid as backup power.
- An electrical grid is used as backup power under certain conditions, for example, in combination with a demand response program where the data center customer relies on its backup diesel generators.

Increase Datacenter Data Transparency to Encourage Capital Market Participation

Energy market participants suffer from the lack of high-quality historical data publicly accessible on datacenter electricity consumption. Datacenter load is only a subcategory of EIA's "commercial load,"

⁴ Tyler H. Norris et al., *Rethinking Load Growth: Regulatory Approaches to Electrification and Demand Growth*, Nicholas Inst. for Energy, Env't & Sustainability, Duke Univ. (Feb. 2024), Available at:

<https://nicholasinstitute.duke.edu/publications/rethinking-load-growth>. (Accessed March 10, 2025).

which is reported monthly. This will be one of the primary consumption categories in the future and should be tracked and understood by the EIA as its own distinct category.

Information creates increased capital participation which leads to lower risk premiums and increased liquidity. Without proper information capital is less willing to participate in the formation of power generation.

Recommendations

Grid interconnection is a meaningful bottleneck for private capital to be deployed in the United States. Our recommendations move to alleviate these issues.

- **Accelerate Grid Interconnection:** Incentivize innovative utility measures (like demand-response and battery storage) to cut down the current 5-year wait time and bridge the power gap.
- **Expand On-Site Generation:** Streamline permitting and develop a national framework to enable grid-connected, large-scale on-site power systems.
- **Enhance Data Transparency:** Establish a dedicated EIA category for datacenter energy use to attract capital and reduce investment risk.

3) Expansion and Adjustment of Existing Controls

It is critical that the United States maintains its leadership and does not export its competitive advantages in AI as the country that created and leads in AI technology; exporting our competitive advantage in this critical technology is helpful to other adversarial nation-states. We believe the current export controls are a step in the correct direction, but have specific recommendations for chips, networking, and semiconductor capital equipment.

AI Chip Exports and Networking

When evaluating an accelerator or GPU, there are three considerations: **Computational power** (measured in Floating Point Operations Per Second, or FLOP/s), **Memory Bandwidth** (measured in TB/s), and **interconnect bandwidth** (measured in GB/s or Gbit/s). Previous export controls have focused on computational power, but this has allowed the optimization of the other two methods of performance. Tens of thousands of Nvidia H20 GPU are purchased legally in China, which can outperform the controlled H100 GPU on some text generation tasks. This is because the initial rules didn't encapsulate holistic performance.

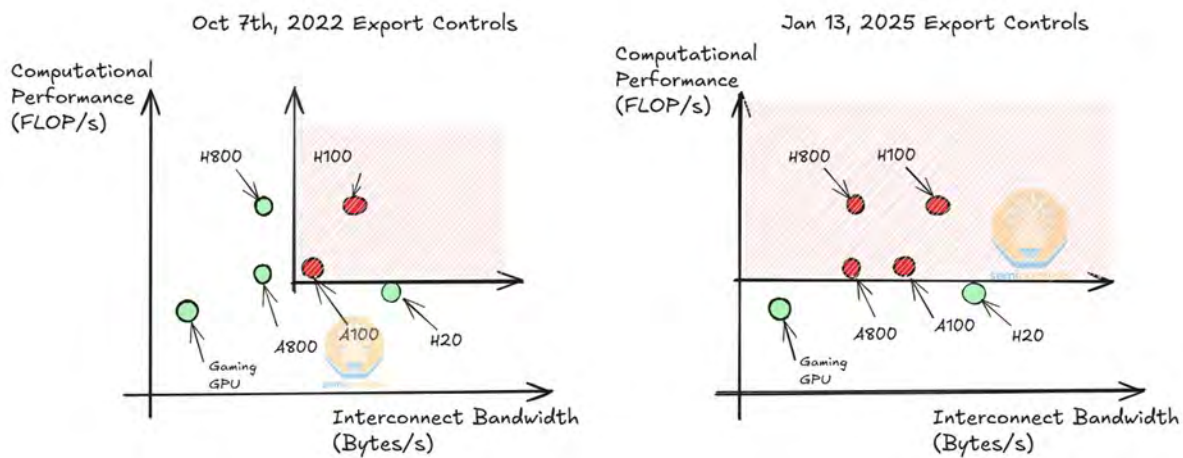


Figure 3 - Controlled and available GPUs under current and previous controls. Source: SemiAnalysis

A single GPU does not exist alone but is part of a cluster of tens of thousands of chips. A slower worker that is more in sync with its collective group can perform just as well as a faster worker that is out of sync. This compares compute (single worker speed) to interconnect (synchronization of GPUs). Two standard technologies connect chips in the “scale up” and “scale out” domains. NVLink is an example of a “scale up” network and Infiniband or Ethernet is an example of a “scale out” network. “Scale up” domains focus on smaller “squads” of chips while “scale out” focuses on a “corps” of chips.

Computational power matters, but a high interconnect bandwidth allows a faster group of chips in aggregate. In other words, even if a chip's specific computational performance is limited, interconnect and memory bandwidths can aggregate many thousands of chips together in a powerful cohesive unit. This renders the original intention of export controls inert.

A more holistic approach to controlling chip exports is imperative. The correct framework considers the entire cluster of chips rather than on an on-chip basis. Export controls should focus on achieving the end goal and not block single technology attributes that can be engineered around. A unified cluster-level framework is a more comprehensive framework that allows for fewer loopholes in the longer term and for the US to maintain leads across more dimensions than just chips with more computational performance.

In addition to the existing restrictions from the Department of Commerce’s January 2025 Interim Final Rule on the Framework for Artificial Intelligence Diffusion (the “Diffusion Rule”), specifically 3A090A.2 and 3A090A.3, we think:

- Memory bandwidth is limited at 4 TB/s per chip
- Total Input/Output (IO) bandwidth is limited at: 9 Tbps bidirectional per chip, including the combined scale up, scale out, and Peripheral Component Interconnect (PCIe).

Any chip that exceeds any limit on these three axes is automatically controlled. Note that this will also include other networking chips that are generally included with the server. As mentioned, this will limit the capability of using more chips and networking switches, which enable better performance as they scale up.

This would **immediately include controls on the Nvidia H20 as a starting point**. We think the H20 should be controlled moving forward. Although it has less computational power than an H100, it has excellent text generation capabilities thanks to its high interconnect bandwidth. Text generation has become an essential part of the new paradigm of “reasoning” where the model thinks through a problem before answering.

Wafer Fabrication Equipment (WFE) or Semicap Tools

Export controls must be stricter further up the semiconductor supply chain. Without restrictions on chip manufacturing equipment and its components, controls on chips themselves are useless—China can simply produce its own advanced chips.

Recent policy has slowed the progress of Chinese chipmakers, starting with EUV restrictions under the first Trump administration and continuing with annual BIS rule updates circa 2022. However, improvements are needed as there are multiple loopholes or workarounds for chip production to continue.

We discussed the need for controls on **subsystems** for semiconductor manufacturing equipment in a report on Chinese sanctions evasions⁵:

“For example, current controls prohibit exports of EUV scanners into China. Yet they do not restrict EUV optics from Zeiss. This makes no sense – ASML and any chipmaker using EUV can tell you that projection optics are the #1 critical component. While it is not practical to control every nut and bolt, certain items obviously must be. Where to draw the line? There are fewer than 100 critical components from specific suppliers that should be restricted, including EUV projection optics, precision mask stages, and radio frequency plasma generators.”

If a Semicap tool is restricted, **its key subsystems must also be restricted**.

Key Semicap tool categories and capabilities must be restricted. There are relatively few categories – closer to 10 than 100 – that must be controlled, not regulated with an arbitrary specification for where “advanced” applications and capabilities begin. Examples include atomic layer deposition (ALD), high aspect ratio etch, and isotropic etch. All are critical for current and future advanced chips, but only tools meeting a certain spec are regulated. This approach makes evasion simple: purchase a tool that is barely

⁵ Dylan Patel et. al, *Fab Whack-A-Mole—Chinese Companies Evading Semiconductor Equipment Export Controls*, SemiAnalysis (Oct. 28, 2024), Available at: <https://semianalysis.com/2024/10/28/fab-whack-a-mole-chinese-companies/>. (Accessed March 12, 2025)

under the allowed spec and modify, retrofit, or push the capabilities of a tool in a way that would be restricted. Instead, the entire category of the tool should be controlled.

Making a complex semiconductor takes thousands of steps, and many of the steps can be replaced or substituted with another category of tools. If we remove one piece, there is often a workaround that can be engineered. Instead, we should remove entire categories of solutions.

Recommendations:

- Work with BIS to amend more holistic export controls, allowing for fewer loopholes to be available for scale-up and scale-out networking.
- Work with BIS to ensure that subsystems of critical machinery are controlled, not just the assembled machine, allowing controls to achieve their targeted goal better.
- Work with BIS to place broader controls on important categories in semiconductor manufacturing, avoiding possibilities of retrofitting or adjusting permitted tools to operate on controlled levels of performance.

4) Enforcement

Export controls are only as effective as their enforcement mechanisms. Due to the lack of enforcement, continuous evasion efforts mean that many wafer-fabricated equipment still reaches Chinese fabs. This must change for controls to be effective and impactful.

The CCP has made domestic chip manufacturing a national priority, yet the BIS is expected to enforce restrictions on them with barely more than 100 agents spread across the entire export controls regime (from chips to automotive to agricultural). This implies that enforcement falls on very few number of people to monitor and enforce billions of dollars in purchases. Controls are only as effective as their enforcement and China has established several lines of smuggling each several hundred million dollars in value.

We have noted multiple ways that export controls get skirted in the past⁶. This sort of observation and enforcement should happen before it reaches the scale of billions of dollars in purchases. Additionally, we believe that meaningful amounts of chips are smuggled into China every year, and enforcement has been lackluster. We outline recommendations that aim to improve enforcement across the board.

Recommendations:

- Provide the BIS with more staffing and resources that it can identify, track, and shut down existing or new lines of smuggling.
 1. This should include highly skilled staff in the semiconductor industry.
 2. Utilize streamlined federal hiring programs to accelerate their involvement.

⁶ Dylan Patel et. al, *Fab Whack-A-Mole—Chinese Companies Evading Semiconductor Equipment Export Controls*, SemiAnalysis (Oct. 28, 2024), Available at: <https://semianalysis.com/2024/10/28/fab-whack-a-mole-chinese-companies/>. (Accessed March 12, 2025)

- Coordinate with BIS to ensure export controls are updated on a monthly cadence to close loopholes quickly, which incurs costs while limiting the advantages gained.
- The Diffusion Rule introduced a 1700 H100 no-license required threshold for orders in Tier 2 countries. While it is essential to ensure that businesses who need only a small number of GPUs to operate procure them with ease, there has been little clarity around managing or enforcing this ruling. Further clarifying details is key to enforcement, working toward efficiently servicing businesses while denying shell companies that pose a smuggling risk.
- Japan and the Netherlands are exempt from Foreign Direct Product Rulings (FDPR) contingent on providing matching controls. However, these matching controls often arrive at a significant lag, allowing suppliers to sell to China. This creates stockpiles of US restricted tools, nullifying the impact of the controls. The US should move to a more unilateral approach to timing controls.
- Task Federal agencies focused on intelligence to work with critical semiconductor companies on building robust internal screening systems to avoid sales to controlled actors through shell corporations.

Conclusion

The future of AI in America is likely the single most important technological inflection this decade and the next. Controlling and effectively deploying this critical technology will have profound and lasting consequences for America's economic competitiveness and national security.

The cost of falling behind in the AI race is severe - the United States would lose its edge in virtually all domains of knowledge work, with cascading effects across military, economic, and technological spheres. To prevent this outcome, we recommend the U.S. implement an integrated strategy focusing on our four key recommendations: establishing robust U.S.-based semiconductor manufacturing, resolving power generation bottlenecks, implementing comprehensive export controls, and enforcing these controls vigorously.

These recommendations form a cohesive framework that is both actionable and achievable. The cost of losing the race is too great to not take seriously, and we urge the federal government to continue to take it seriously and adopt these recommendations as foundational elements of the AI Action Plan.

SemiAnalysis believes a strong American AI industry is critical for the future and are always open for continued dialogue. We can be reached at [REDACTED]

About SemiAnalysis: SemiAnalysis is the best-of-breed Semiconductor and AI Research boutique. Our coverage spans the entire supply chain, from semiconductor fabrication essentials to cutting-edge AI models, software, and infrastructure.